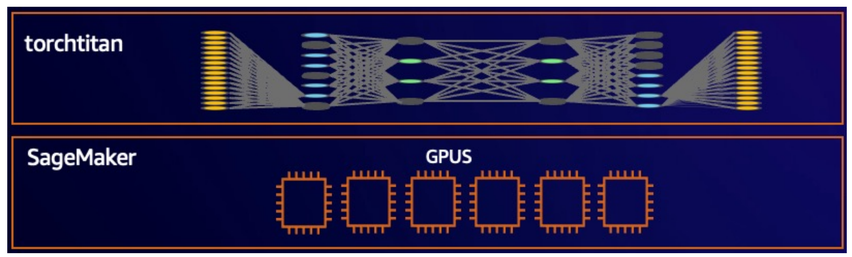
Efficient Pre-training of Llama3-like architectures using torchtitan library on Amazon SageMaker

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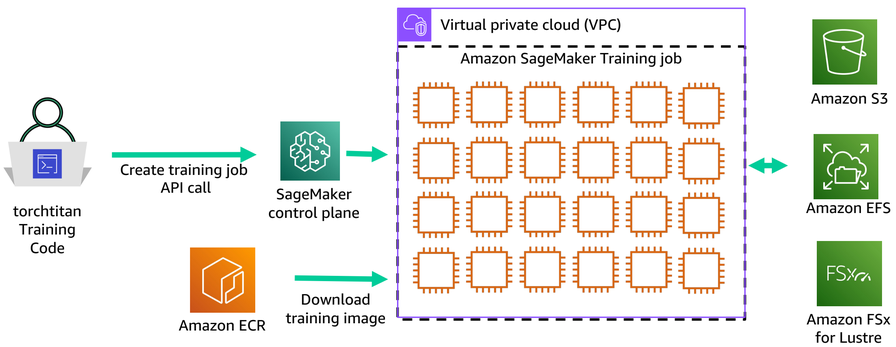
Abstract

AWS and Meta's PyTorch teams demonstrate scalable and memory-efficient pre-training of Llama-3-like architectures using torchtitan library on Amazon SageMaker.



Amazon SageMaker Training Jobs

SageMaker is a fully managed machine learning (ML) service that can help you build, train, and deploy ML models into a production-ready hosted environment



- **Managed on-demand capacity**, enabling granular and cost-effective GPU utilization.
- **Resiliency through automated GPU health checks** and node-level failure remediations.
- **Streamlined data ingestion** during training by automating I/O operations with AWS object and file system storage services.

Meta's torchtitan

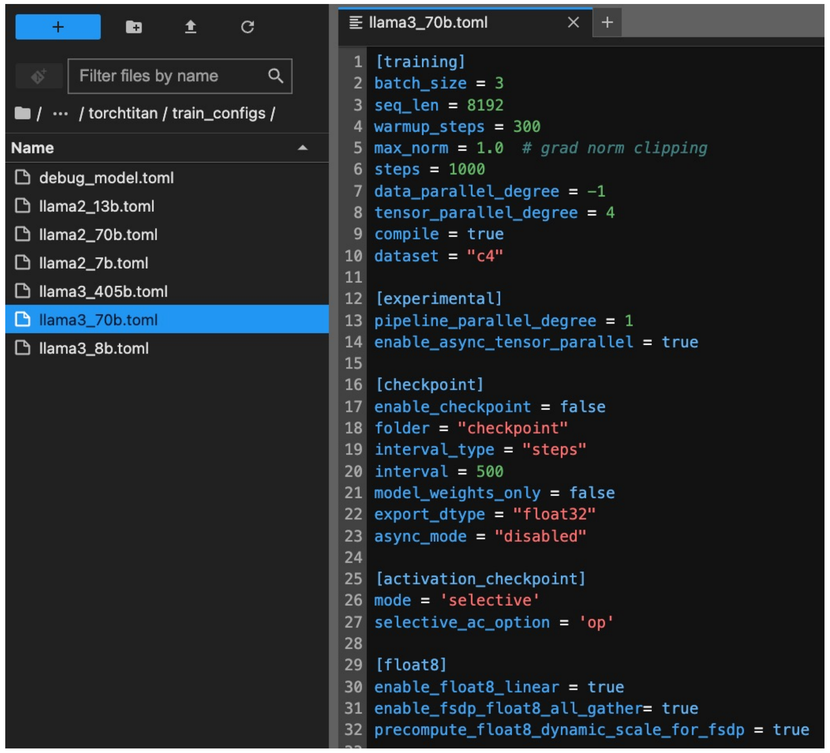
https://github.com/pytorch/torchtitan

torchtitan is a PyTorch-based package for large-scale language model training, showcasing the latest distributed training features in a clean and minimal codebase.

Key features:

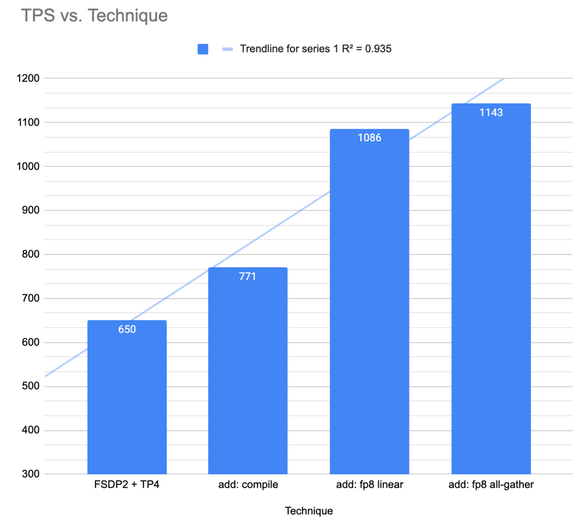
- FSDP2 support through DTensors
- torch.compile support
- FP8 support for accelerated linear layer computations
- FP8 all_gather for faster parameter communications
- Async Tensor parallelism
- Selective layer and operator activation checkpointing

All options easily configured via toml files:

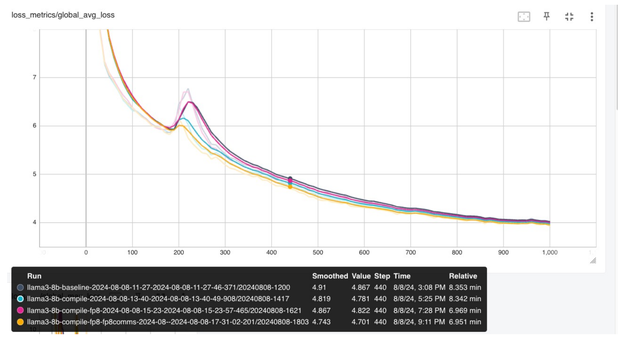


Experiments

Training Llama-3 70B model with torchtitan on 128 H100 GPUs with incremental optimizations:



Loss curves for Llama-3 8B pre-training:



Conclusion:

- **Efficient pre-training:** Torchtitan enables efficient Llama-3 pre-training with FSDP2, achieving a **76%** speedup over the baseline for Llama-3 70B pre-training
- **Scalable infrastructure:** Amazon SageMaker provides a managed, cost effective and resilient training environment, through on-demand GPU capacity, spot instances and GPU health checks.